



Capture of Freely-Floating Planetesimals into Protoplanetary Disks

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Introduction

In 2017, an object was observed passing through our solar system at an observatory in Hawaii (Meech et al., 2017). The object, known as Oumuamua, was the first interstellar planetesimal ever observed directly. Experts were able to determine its interstellar origins based on its hyperbolic trajectory. The discovery of a free-floating planetesimal in our solar system provided evidence to support the long-sSpeculated idea that there are many free-floating planetesimals in the interstellar medium. This abundance of planetesimals could be an important factor in the process of planet formation, as they could be captured by protostars and and grow via accretion of solid particles from the protoplanetary disk. To better understand this process, we will be studying the impact of gravitational potential on the trajectory and capture rate of planetesimals in collapsing protostars.

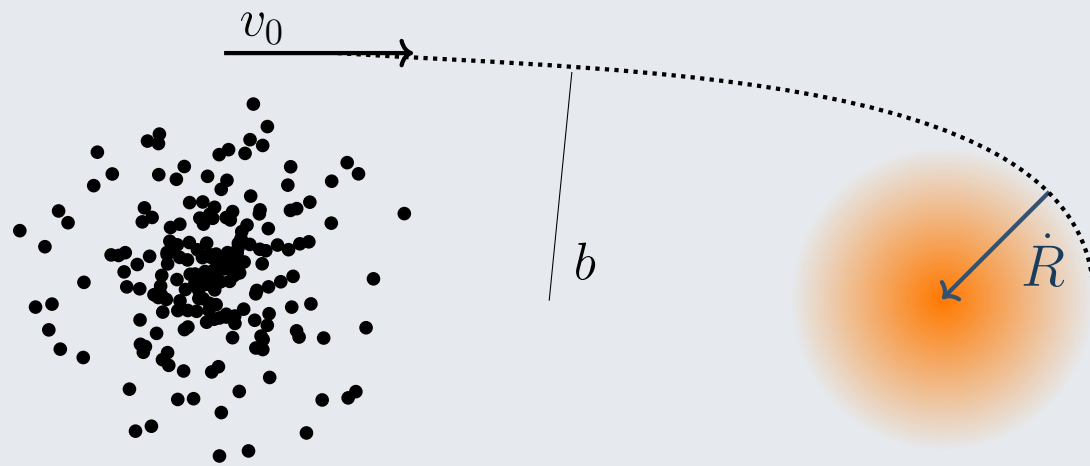


Figure 1: Schematic Representation of planetesimals approaching proto-star at initial velocity v_0 , which is collapsing at a rate $\dot{R}$.

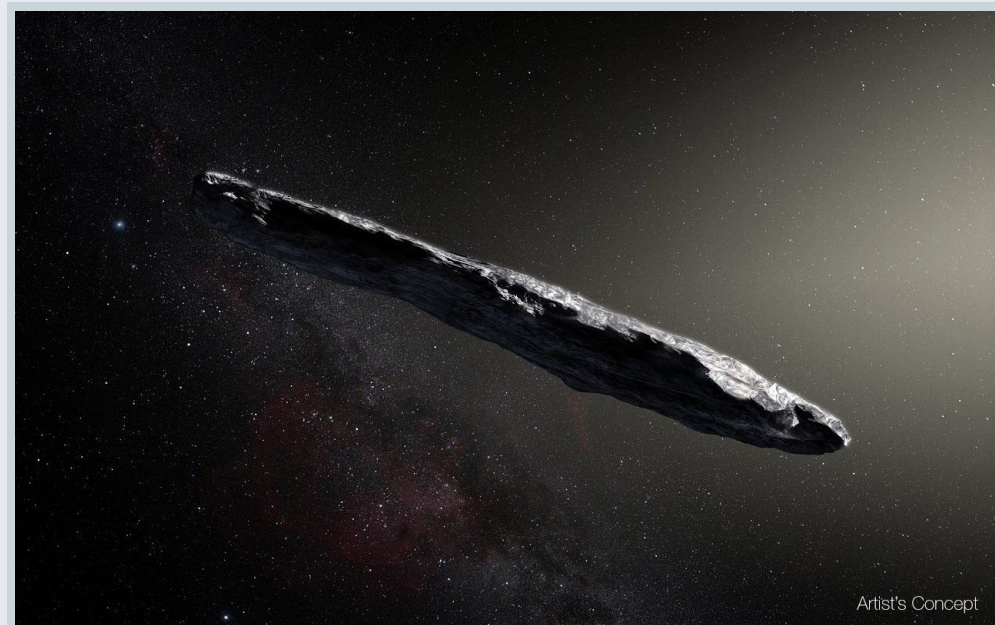
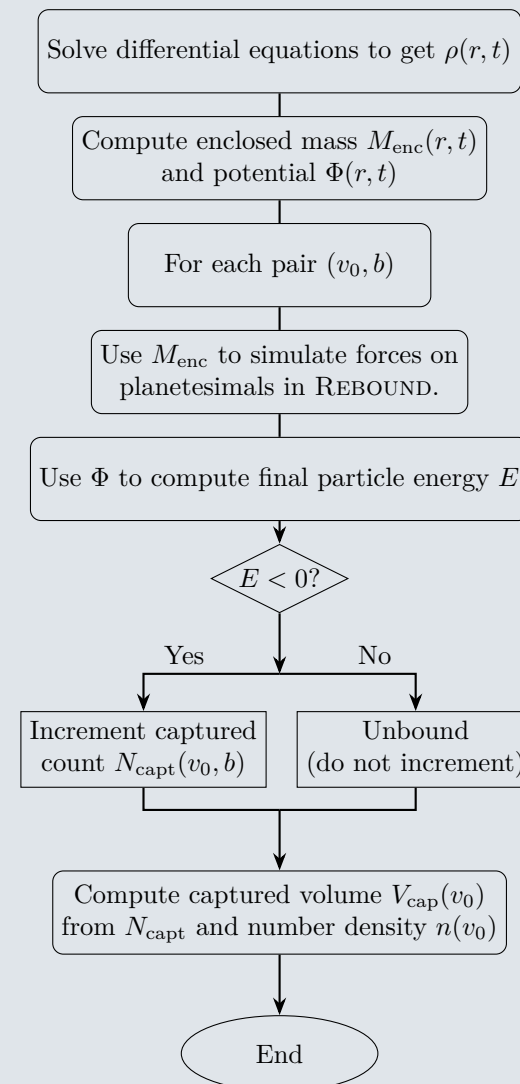


Figure 2: NASA. (2017, November 20).

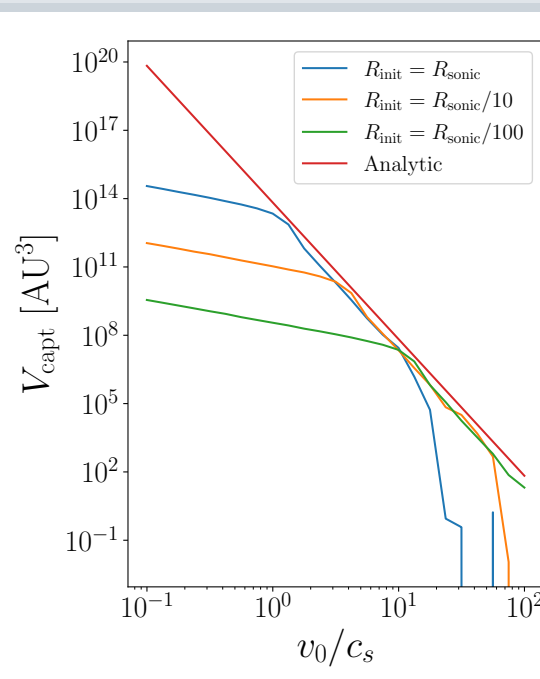
Methodology



The flowchart on the left shows the general structure of the main simulation, where each simulation is run for some initial velocity and range of impact parameters. We numerically solve a set of coupled differential equations to obtain an expression for $\rho(r, t)$, which allows us to implement a function in REBOUND(Rein & Liu, 2012) which simulates the force of the collapsing proto-star on the planetesimals. After running the simulations for some pairs of initial velocities (v_0) and impact parameters (b_i), we can check if $E < 0$ to assign a capture number to each simulation, and calculate the capture volume as

$$V_{\text{cap}} = \sum_i \pi \frac{N_{\text{capt}}^i}{N_{\text{static}}^i} (b_{i+1}^2 - b_i^2) v_0.$$

Homogeneous Model



In the homogeneous model, the rate at which the proto-star collapses is modeled as

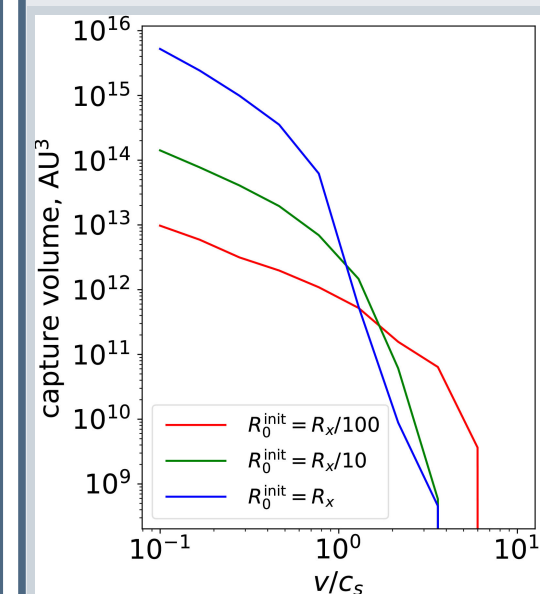
$$\dot{R} = -\alpha \sqrt{\frac{2GM}{R}}.$$

Where here, α is a number between 0 and 1, therefore the cloud collapses at some fraction of the free-fall rate. The homogeneous case provides a test case to develop the simulation and validate against the analytic approximation,

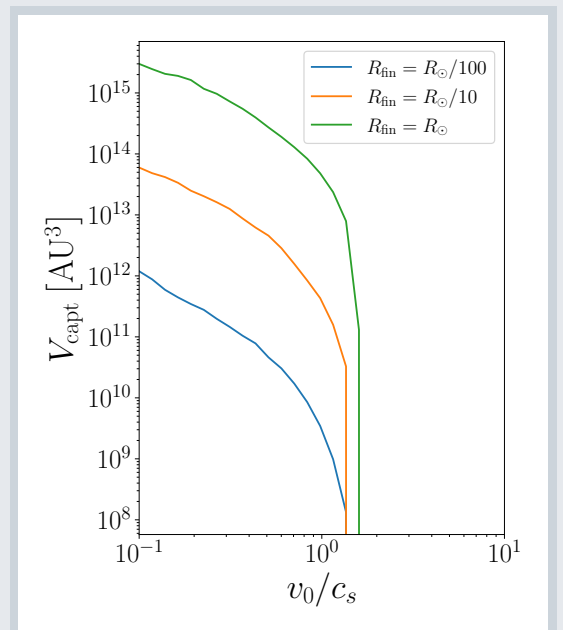
$$V_{\text{capt}}(v_0) = 40.8\alpha^{3/2} \left(\frac{GM}{v_0^2} \right)^3.$$

We have observed that the capture rate for the Larson (1969) (before the collapse of the proto-star) and Shu (1977) (after the collapse of the proto-star) cases behave qualitatively differently than the homogeneous case, with planetesimals going a few times the sound speed not being captured.

Larson Model



Shu Model



Discussion

Although planetesimals can be captured through gas drag, as shown in Grishin et al. (2019) and Moro-Martín & Norman (2022), this mechanism is less effective for larger bodies. The capture of such large planetesimals is particularly significant, since they can grow into planet-sized bodies by rapidly accreting centimeter-scale dust grains from the disk in a process known as pebble accretion (Lambrechts & Johansen, 2012). In future work, we plan to combine our results with a realistic size distribution for interstellar planetesimals. This will allow us to assess whether enough large planetesimals can be captured to meaningfully accelerate planet formation around young stars.

References

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